

PHISING EMAIL DETECTION & AWARENESS REPORT

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TASK: FUTURE_CS_O2



INTRODUCTION

Phishing is one of the most common cyber threats facing individuals and businesses today. Attackers make deceptive emails that impersonate trusted organisations in order to trick users into clicking malicious links, surrendering passwords or downloading harmful files.

This report presents the analysis of three phishing email samples collected from a public research repo. Each email has been examined for phishing indicators, classified by risk and used to build practical awareness guidelines for everyday users

METHODOLOGY & TOOLS

Analysis Approach: Each email was examined in its raw .eml format with headers and body content inspected manually.

Tools Used:

Raw .eml file inspection (Notepad / text editor)

Google Apps Message Header Analyser (toolbox.googleapps.com)

MXToolBox Email Header Analyser (mxtoolbox.com)

Browser tools for safe URL inspection

Classification System:

-  Safe – No indicators of phishing
-  Suspicious – 1 to 2 red flags present
-  Phishing – 3 or more red flags confirmed

SAMPLE ANALYSIS – SAMPLE 1 File:sample-1.eml

Type: Fake bank loyalty Rewards

Classification: 🚨 Phishing

Email Details:

> **From:** banco.bradesco@atendimento.com.br

> **Return-Path:** root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06

> **Subject:** Your 92,990 Livelo points expire today!

> **Malicious Link:** blog1seguimentmydomaine2bra.me

Phishing Indicators Identified:

Indicator	Detail
Spoofed sender domain	Claims to be Bradesco Bank but sent from a random VPS server
Domain mismatch	From address and Return-Path are completely different
Failed authentication	DKIM unsigned,DMARC failed,SPF error
Urgency tactic	Points “expire TODAY” pressures user to act immediately
Fake link	Redemption link has no connection to Bradesco

How the Attack Works: The attackers impersonates a well known Brazilian bank and tells and recipient their reward points are about to expire. The urgency pushes the user to click a fake link which likely leads to credentials being taken to steal banking login details

Evidence:

MessageId	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Created at:	9/19/2023, 8:35:49 PM GMT+2 (Delivered after 56 sec)
From:	BANCO DO BRADESCO LIVELO<banco.bradesco@atendimento.com.br>
To:	phishing@pot
Subject:	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
SPF:	temperror with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	temperror Learn more

SAMPLE ANALYSIS – SAMPLE 2 File:sample-10.eml

Type: Fake Microsoft Account

Classification: 🚨 Phishing

Email Details:

> **From:** no-reply@access-accsecurity.com

> **Return-Path:** bounce@thcultarfdes.co.uk

> **Reply-To:** sotrecognizd@gmail.com

> **Subject:** Microsoft account unusual signin activity

Phishing Indicators Identified:

Indicator	Detail
Fake Microsoft domain	Real Microsoft emails come from @ mircosoft.com not @ access-accsecurity.com
Three Domain mismatch	From,Return-Path and Reply-To are all different unrelated domains
Gmail Reply-To	Microsoft would never use a personal gmail as a reply address
Fear tactic	“Unusual signin activity” creates panic to force a quick reaction
No authentication	SPF none,DKIM none,DMARC error – email completely unverified

How the Attack Works: The attacker impersonates a well known brazilian bank and tells the recipient their reward points are about to expired. The urgency pushes the user to click a fake link, Which likely leads to a credentials page designed to steal banking login details.

Evidence:

MessageId	032672b4-77ca-42f8-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com
Created at:	9/8/2023, 7:47:04 AM GMT+2 (Delivered after)
From:	Microsoft account team _<no-reply@access-accsecurity.com>
To:	phishing@pot
Subject:	Microsoft account unusual signin activity
SPF:	none Learn more
DKIM:	none Learn more
DMARC:	permerror Learn more

SAMPLE ANALYSIS – SAMPLE 3 File:sample-100.eml

Type:Fake Solar Panel Deal

Classification: 🚨 Phishing

Email Details:

> **From:**zonnepaneel@appjj.serenitepure.fr

> **Return-Path:**return@dturm.de

> **Reply-To:** news@aichakandisha.com

> **Subject:** Solar panels at a good price

Phishing Indicators Identified:

Indicator	Detail
Complete domain mismatch	From,Return-Path and Reply-To are three entirely different domains
Unrelated sender domain	serenitepure.fr has not connection to solar panels
Geographic inconsistency	Sent from a german server using a French domain targeting Dutch speakers
Zero authentication	SPF none,DKIM none,DMARC none–worst possible score
Suspicious routing	Email routed through multiple unrelated servers before delivery

How the Attack Works:This is a lure based phishing email offering an attractive deal on solar panels.The goal is to get the person to click a link or respond ,leading them to either a fake purchase page that steals payment details or a malware download

Evidence:

MessageId	0.0.0.0.1D8EF409A5C12CE.37AA@dturm.de
Created at:	11/3/2022, 6:56:15 AM GMT+2 (Delivered after)
From:	"Zonnepanelen installateur" <zonnepaneel@appjj.serenitepure.fr>
To:	phishing@pot
Subject:	 Zonnepanelen voor een goede prijs
SPF:	none Learn more
DKIM:	none Learn more
DMARC:	none Learn more

PHISHING INDICATORS SUMMARY

Indicator	Sample 1	Sample 2	Sample 3
Spoofed sender domain	✓	✓	✓
Domain mismatch(Return-Path)	✓	✓	✓
Failed SPF/DKIM/DMARC	✓	✓	✓
Urgency or fear language	✓	✓	x
Suspicious links	✓	✓	✓
Generic or no greeting	✓	✓	✓
Total Red Flags	5	5	4
Classification	 Phishing	 Phishing	 Phishing

PREVENTION GUIDELINES

How to Spot a Phishing Email – User Guide

Before clicking any link in an email, ask yourself these questions:

- > Does the sender's email domain match the real company? (e.g. [@microsoft.com](mailto:info@microsoft.com) not [@access-accsecurity.com](mailto:info@access-accsecurity.com))
- > Is the email pressuring you to act urgently or threatening negative consequences?
- > Does the greeting use your real name or just "Dear User" / "Dear Customers"?
- > Does Hovering over the link show a sus or unrelated URL?
- > Were you expecting this email?

If the answers to any of these is no or unsure – do not click.

DO's:

- >  Always check the sender's full email address, not just the display name
- >  Hover over links before clicking to preview the real URL
- >  Go directly to a website by typing the URL rather than clicking email links
- >  Report suspicious emails to your IT or security team
- >  Use multi-factor authentication on all important accounts
- >  Keep your email provider's spam filter turned on

DON'Ts:

- >  Don't click links in emails that create urgency or fear
- >  Don't enter passwords or personal details from a link in an email
- >  Don't download attachments from unknown senders
- >  Don't trust an email just because it looks professional or uses a company logo
- >  Don't reply to suspicious emails, even to unsubscribe

CONCLUSION

All three email samples analysed in this report were confirmed as phishing attempts. Each used a combination of domain spoofing, authentication failures and psychological tactics to deceive recipients. The most consistent indicators across all samples were sender domain mismatches and the complete absence of email authentication (SPF, DKIM, DMARC).

User awareness is the most effective defence against phishing. By learning to inspect sender addresses, verify links before clicking, and question urgency-driven emails, individuals and organisations can significantly reduce their risk of falling victim to these attacks.